Scalability & Future Plan

Date	29 October 2026
Team ID	LITVIP2026TMID00420
Project Name	EV Market Analysis Using Tableau
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Static historical dataset only, no real-time data integration	Cannot show live EV market trends and live sales	High
2	Dashboard built on Tableau Public with limited data handling capacity	Performance slow when large EV dataset (millions of records) loaded	Medium
3	Only descriptive analysis, no AI/ML based prediction model	Cannot forecast future EV sales growth and demand	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Only 10-20 users can view locally on Tableau Public	Deploy dashboard on Tableau Cloud / Server to handle 1000+ concurrent users
2	Data Storage	Limited to local Excel / CSV files	Migrate data to Cloud database like Snowflake / BigQuery for large EV datasets
3	Performance	Slow loading when applying region/category	Use Tableau Extract (.hyper) and data aggregation for fast performance

		filters on large data	
4	Security	Public dashboard, no user authentication	implement role-based access control and secure private embedding

Future Roadmap:

Phase	Planned Feature / Enhancement Target Timeline Expected Impact

Phase 1

Planned Feature: Real-time EV sales data integration via APITarget Timeline: 1 MonthExpected Impact: Live market trends and accurate insights

Phase 2	Planned Feature: Real-time EV sales data integration via APITarget Timeline: 1 MonthExpected Impact: Live market trends and accurate insights
Phase 3	Planned Feature: Mobile responsive dashboard and improved UITarget Timeline: 3-4 MonthsExpected Impact: Wider user reach and better user experience
Phase 4	Planned Feature: Expand analysis to Global EV charging infra & government policiesTarget Timeline: 6 MonthsExpected Impact: Complete EV ecosystem analysis